

Designing Accurate Data Entry Procedures

CSE 4407

Md. Bakhtiar Hasan

Assistant Professor
Department of Computer Science and Engineering
Islamic University of Technology

August 20, 2025



Effective Coding

- **What:** The process of putting ambiguous or cumbersome data into **short, easily entered digits or letters**
- **Why:** Being eloquent but succinct
 - **Efficiency:** Less time to enter, fewer items to enter → E.g., LAX vs. Los Angeles International Airport
 - **Accuracy:** Fewer keystrokes = fewer chances for typos
 - **Storage:** Saves valuable memory and storage space
 - **Processing:** Helps in sorting and data transmission later

The Five Purposes of Coding

- **Keeping Track** of something → E.g., an order
- **Classifying** information → E.g., student year
- **Concealing** information → E.g., wholesale price
- **Revealing** information → E.g., product details on a tag
- **Requesting** appropriate action → E.g., 'Delete' command

Purpose 1: Keeping Track of Something

- **Goal:** Identify a person, place, or thing uniquely
- The code itself does not need to describe the item
- **Two Common Types**
 - Simple Sequence Codes
 - Alphabetic Derivation Codes

Simple Sequence Codes

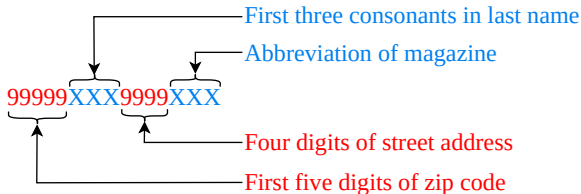
- A number assigned to something just to number it
- No relation to data itself
- **Example:** A furniture manufacturer's order numbers

Order #	Product	Customer
5676	Rocking Chair/with Bamboo	Feng Xu
5677	Dining Room Chair/Upholstered	Millie Monice
5678	Love Seat/Upholstered	J. & D. Yang
5679	Child's Rocking Chair/Decals	Chelsea Kennard

- **Advantages**
 - Eliminates assigning the same number twice
 - Gives an approximation of when the order was received
- **Best Used:** Order of processing matters (e.g., first-come, first-served)

Alphabetic Derivation Codes

- Code is derived from the data itself
- Avoids simple sequences that reveal volume
- Example: Magazine subscription account number



- Advantages
 - Identify an account
 - Pre-sorting for bulk mail (using zip code)
- Disadvantages
 - What if the name is short on consonants? → Requires a backup plan
 - What happens if the customer moves? → Major database headache

Purpose 2: Classifying Information

- **Goal:** Distinguish among classes or groups of items
- **Rule:** Classes must be *mutually exclusive*
 - A student cannot be both a Freshman and a Sophomore
 - Overlapping classes create confusion
- **Two Common Types**
 - Classification Codes
 - Block Sequence Codes

Classification Codes

- Uses a single letter or number to represent a group
- **Example:** Coding Tax-Deductible Item

Code	Tax-Deductible Item
I	Interest Payments
M	M edical Payments
T	T axes
C	C ontributions
D	D ues
S	S upplies
B	Sub b scriptions
P	Comp b uter
N	I nsurance

- **Problem:** What happens when you run out of unique first letters? → Force it?

Block Sequence Codes

- An extension of the sequence code
- Assigns *blocks* of numbers to specific classes
- Within a block, numbers are sequential
- **Example:** Software Inventory

Code	Name of Software Package	Type
100	Apple Safari	Browser
101	Mozilla Firefox	
.		
.		
200	Microsoft Access	Database
201	MySQL	
.		

- **Advantage:** Combines grouping with the simplicity of sequencing

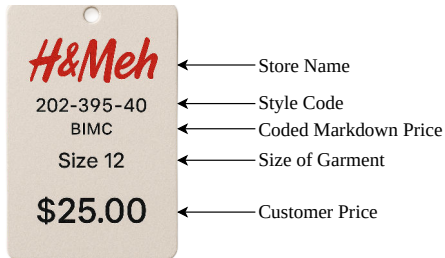
Purpose 3: Concealing Information

- **Goal:** Disguise information you do not want others to know
- **Why**
 - Protecting sensitive data
 - Ensuring security for online transactions
- **Method:** Cipher Code

Cipher Codes

- A simple substitution code → Letter for letter, letter for number
- **Example:** Encoding markdown prices

Code	Meaning
B	1
L	2
E	3
A	4
C	5
H	6
M	7
I	8
N	9
D	0



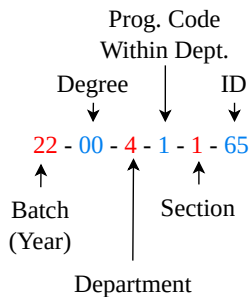
- Regular Price: \$25.00
- Markdown Ticket Encoded BIMC: \$18.75

Purpose 4: Revealing Information

- **Goal:** Reveal details to those who know the code's structure
- Makes data entry more meaningful and helps staff
- **Example:** A clothing price tag that codes department, style, color, and size
- **Types**
 - Significant-Digit Subset Codes
 - Mnemonic Codes
 - Unicode

Significant-Digit Subset Codes

- A long code made up of smaller, meaningful sub-codes
- Each “subset” of digits has its own meaning
- **Example:** Student ID of IUT students



- **Advantage:** Allows for easy searching and grouping

Mnemonic Codes

- A human memory aid
- Uses letters and symbols to create and easily recognizable code
- **Goal:** Make the code easy to remember and understand
- **Example:** City hospital codes

Code	City Hospitals
BGH	B uffalo G eneral H ospital
ROS	R oswell Park Memorial Institute
KEN	K enmore Mercy
DEA	D eaconess Hospitall
SIS	S isters of Charity
STF	S aint F rancis Hospital
STJ	S aint J oseph's Hospital
OLV	O ur L ady of V ictory Hospital

- Reduces error by being intuitive

Unicode

- **Problem**
 - Standard keyboards only support Latin characters
 - What about Greek, Japanese, Chinese, Hebrew?
- **Solution:** Unicode → Universal Character Encoding
 - A standard that includes all language symbols
 - Can represent over 65,000 characters
- **Format:** `&#xnnnn;`
- **Example**

Character	こ	-	ん	-	に	-	ち	-	は
Hiragana	ko	-	n	-	ni	-	chi	-	ha*
Unicode	3053	-	3093	-	306B	-	3061	-	306F

Purpose 5: Requesting Appropriate Action

- **Goal:** Instruct the computer or a decision maker on what action to take
- Typically uses a sequence or mnemonic code
- **Example:** HTTP status codes

Code	Meaning
200	OK
201	Created
301	Moved Permanently
302	Found (Redirect)
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Internal Server Error
503	Service Unavailable

General Guidelines for Coding

- Keep codes **concise**
- Keep codes **stable**
- Make codes **unique**
- Allow codes to be **sortable**
- Avoid **confusing** codes
- Keep codes **uniform**
- Allow for **modification**
- Make codes **meaningful**

Core Principles for Effective Coding Systems

- **Concise**
 - Shorter is better → Fewer keystrokes, less memory
 - Break up long codes: 244-1139
- **Stable**
 - Should not change
 - Don't include volatile data in code
- **Unique:** No two items should have the same code
- **Uniform**
 - Be consistent
 - Don't mix XXX-99 and XX-9999
- **Allow Modification**
 - The business will grow
 - Plan for new categories, more customers, etc.

Designing Sortable Codes for Efficient Organization

- **Sortable:** Must sort correctly

Incorrect Sorting Using MMM-DD-YYYY	Incorrect Sorting Using MM-DD-YYYY	Incorrect Sorting (Y2K Problem) YY-MM-DD	Correct Sorting Using YYYY-MM-DD
Dec-25-1998	06-04-1998	00-06-11	1997-06-12
Dec-31-1997	06-11-2000	97-06-12	1997-12-31
Jul-04-1999	06-12-1997	97-12-31	1998-06-04
Jun-04-1998	07-04-1999	98-06-04	1998-10-24
Jun-11-2000	10-24-1998	98-10-24	1998-12-25
Jun-12-1997	12-25-1998	98-12-25	1999-07-04
Oct-24-1998	12-31-1997	99-07-04	2000-06-11

Clarity and Meaning in Code Design

- **Avoid Confusing Codes:** Don't use characters that look alike

Handwritten Code	Actual Code	Problem
L85 4M4	L8S 4M4	S looks like a 5
T3A ZES	T3A 2E5	2 looks like a Z 5 looks like an S
LOS IJO	L0S 1JO	Zero looks like an O 1 looks like an I

- **Meaningful**
 - Codes should make sense to users
 - Unless you are intentionally concealing info

Deciding What to Capture

- **Key Idea:** The quality of your output depends entirely on the **quality of your input**
- **Decision**
 - Involves analysts and users
 - Capturing, storing, and retrieving data is costly
- **Types of Data**
 - **Changeable Data**
 - Varies with every transaction
 - The **quantity of pages in an office supply order**
 - **Differentiation Data**
 - **Uniquely identifies** an item or person
 - A **patient's Social Security Number and last name**
- **Goal:** Capture only what is necessary for the transaction

Let the Computer Do the Rest!

- Avoid needless data entry
- Computers Excel at
 - Handling repetitive tasks automatically
 - Recording the time and date of transactions
 - Calculating new values from input
 - Storing and retrieving data on demand
- The “Enter Data Once” Principle
 - Online Computer Library Center (OCLC): A library item is cataloged once, then shared with thousands of libraries
 - Frequent-Flyer Miles: User enters a flight number, then the system calculates and updates the mileage balance

Data Capture in Modern GUIs

- **Drop-down lists** are a key tool for accurate data capture
- **How It Works**
 - The list **displays both the Code and its Meaning**
 - User sees meaningful text
 - System stores the efficient code
- **Benefits**
 - Prevents users from guessing codes
 - **Makes it impossible to enter an invalid code**

Number	Customer Name	Account Status	Account Type	Salesperson	Street
55448	Al Beback	0	V	845	404 Notafound St.
31889	Anita Bath	0	C		d.
97220	Bill Board	0	D		
64680	Beau Tox	0	E		it Rd.
80173	Chris P. Bacon	0	G		
80568	Drew Peacock	1	T		ive
74480	Ella Vator	0	V		
11781	Gail Forcewind	2	V	086	666 Deja Vu Ave.
05044	Justin Time	2	G	652	12 Alibi Room Rd.
94374	Lois Price	1	G	447	0 Clue Crescent
71913	Manny Jahs	2	D	343	86 O'Clock Shadow St.
14307	Neil Down	1	C	253	404 Error Ln.
27762	Paige Turner	1	E	411	10 Punderful Place
37526	Rick O'Shea	0	V	677	88 Witty Bitty Blvd.
20882	Sal Ami	2	C	741	1 Eye C U St.
07804	Sue Flay	1	D	503	9 Lives Alley
94478	Terry Aki	1	D	803	13 Laffy Taffy Terrace
37455	Will Power	2	E	435	32 Bit Byte Blvd.
40178	Warren Peace	1	T	002	77 Sigh Avenue
31834	Yul B. Sorry	2	C	026	22 Giggabyte Grove

System Flow and Form Design

- **Avoiding Bottlenecks**
 - When data entry is slowed because the system cannot handle the volume
 - Ensure your system has enough capacity for peak loads
- **Avoiding Extra Steps**
 - Fewer steps = fewer chance of error
 - Online, real-time systems are great for this
- **Good Source**
 - Form must be well-designed
 - Clear form leads to clear data entry

Choosing a Data-Entry Method

- Many ways to get data into a system
- Main Contenders
 - Keyboards
 - Bar Codes
 - RFID and NFC
 - Optical Character Recognition
 - Magnetic Ink Character Recognition
 - Mark-Sense Form

Bar Codes: Code for Codes

- Metacodes that encode numbers and letters
- Main Types
 - 1D (Linear) Bar Codes: Standard UPC codes on products
 - 2D (Matrix) Bar Codes: Hold more information in less space



(01)08562600027938



QR Codes: More Than Just a Link

- Free to use, identifiable by the 3 corner squares
- Imaginative Uses
 - Go to a website
 - Initiate an email or phone call
 - Get a digital coupon
 - View a promotional video
 - Register for an event
 - Reserve a room
- Pros
 - Everyone has a reader
 - High accuracy
 - Free and easy to generate
- Cons
 - Can be tampered with
 - Security risk

RFID and NFC: Hands-Free Data Capture

- Radio Frequency Identification (RFID)
 - Uses tags (chips + antenna) and a reader
 - Passive Tags: No power, cheap, used for inventory
 - Active Tags: Powered, reliable, used for logistics
 - Con: Major privacy concerns
- Near Field Communication (NFC)
 - Builds on RFID, allows two-way communication
 - Very short range, considered secure
 - Use Case: Contactless payments

Recognition Technologies

- Optical Character Recognition (OCR)

- Reads printed or handwritten text from a source document
- Can speed up data entry by 60-90%
- Great for digitizing faxes or paper documents into editable text

- Magnetic Ink Character Recognition (MICR)

- Reads the special magnetic ink characters on the bottom of checks
- High-speed, reliable, and secure

- Mark-Sense Forms

- A scanner senses where marks are made on a form
- Used for multiple-choice test, surveys
- **Con:** Prone to errors from stray marks and confused users

এইচএসসি পরীক্ষা [২০১৭]
সেবারিক অফিসার উত্তরণের

উত্তরণের নির্ধারিত স্থান ব্যতিত কোন অব্যাহতি পাওয়া দেয়া বা কোন কিছু সেবা যাবে না
 অবশ্যই কোনো বলা-পয়েন্ট কাম দিয়ে নৃত্র ভর্তি করতে হবে।

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(বি.প্র. কোন একজন লাইন সোল্ড সেবা নিয়েছে)

নিয়মানুযায়ী

- বুদ্ধিমত্তা যাচাইকরণ প্রক্রিয়ায় অন্তর্ভুক্ত করা হবে যাদের একজনকে ডায়েরির ফর্মগুলি দেখা না যায়।
- পড়তে পড়িয়ে *
স্থল পড়িয়ে *
- বুদ্ধিমত্তা যাচাইকরণ অবশ্যই বল-পয়েন্ট কাম দিয়ে করতে করতে হবে।
- উত্তরণের কোন অব্যাহতি পাওয়া যাবে না।
- উত্তরণকারীদের কোন অব্যাহতি নেই।
- পরীক্ষার পরীক্ষার্থী ও উপস্থিতির উত্তরণের ক্ষমতা রয়েছে।
- স্টেট কোড বা নিয়ন্ত্রণ / বৃত্ত জরিপ বা করলে উত্তরণ পরে হাল্কা হবে।

Input Validation

- Core Idea

- Assume errors will happen
- Catch them before they contaminate the database

- Types

- Validating Input Transaction

- Submitting the wrong data
- Data submitted by an unauthorized person
- Asking the system to perform an unacceptable function

- Validating Input Data

- Missing data
- Incorrect field length
- Unacceptable composition
- Out of range
- Invalid

Level 1: Validating the Transaction

- Is the overall submission legitimate?
- Common Checks
 - Submitting the Wrong Data?
 - Example: Trying to input a patient's medical record number into the payroll system
 - The system should reject it
 - Submitted by an Unauthorized Person?
 - Example: A junior clerk trying to approve a manager's expense report
 - Access should be denied
 - Requesting an Unacceptable Function?
 - Example: Trying to DELETE a customer record when your role only allows you to VIEW them
 - The system should reject it

Level 2: Validating the Data - Basics

- Checking each individual data field
- Common Checks
 - Presence Check
 - Is required data missing?
 - Example: A new user registration is invalid without an email address
 - Length Check
 - Is the data the correct length?
 - Example: A US State Code must be 2 characters, not 1 or 3
 - Composition Check
 - Is the data of the right class?
 - Example: A field for Quantity must be a number

Level 2: Validating the Data - Sanity and Contextual Checks

- Does the data make sense in context?
- Common Checks
 - Range/Reasonableness Check
 - Does the data fall within acceptable limits?
 - Example (Range): The Month field must be between 1 and 12
 - Example (Reasonableness): A birth year of 1790 for a new employee is probably not reasonable
 - Invalid Values Check
 - Does the data match a predefined list of allowed values?
 - Example: A field for Payment Type can only be Cash, Credit, or Check, NOT V-Bucks
 - Cross-Reference Check
 - Do related fields make sense together?
 - Example: Sale Price must be greater than Cost Price

The Check Digit: A Self-Validating Code

- **Goal:** Catch common data entry errors for numbers
 - **Single Digit Miskey:** 2441139 → 2451139
 - **Transposition Error:** 2441139 → 2441193
- **How?**
 - Start with the original number
 - Use a mathematical formula to calculate a single extra digit
 - Append it to the end
- **Benefit**
 - When the number is entered, the system re-runs the calculation
 - If the results do not match, the **number is invalid**

Case Study: Verifying a Credit Card

- Multi-layered process
 - Length Check
 - Is it the right number of digits?
 - Example: Amex 15 digits, Visa/MasterCard 16 digits
 - Prefix Check
 - Do the first few digits match the card type?
 - Example: Amex starts with 34/37, Visa 4, and MasterCard 51-55
 - Check Digit Validation: Using the Luhn Formula
 - Extra Security: Expiration Data and the CVV code
- Each layer adds more certainty that the card number is valid

The Luhn Formula (Simplified)

- Algorithm used to validate credit cards
- **Example:** 3141592653
 - Double every other digit, starting from the second-to-last $\rightarrow$ 6 1 8 1 10 9 4 6 10 3
 - If doubling a digit gives a 2-digit number, add its digits together $\rightarrow$ 6 1 8 1 1 9 4 6 1 3
 - Sum all the digits of the new number $\rightarrow$ 40
 - If the total ends in 0, the number is valid!
- **Limitation:** Catches most single-digit errors, but not all transposition errors

The Validation Process and Advanced Tools

- Order of Operations
 - Check for missing data
 - Then, check syntax (length, composition)
 - Only then, check semantics (range, values, check digits)
 - Finally, perform cross-reference checks
- Advanced Tools: Regular Expressions (Regex)
 - Power way to define complex text patterns
 - Example: `[A-Za-z0-9]\w2,@[A-Za-z0-9]3,.[A-Za-z]3/` for email
 - Careful: Google RFC822/RFC2822 OR Checkout [Falsehoods about Email Addresses](#)

Ecommerce and Data Accuracy: A Digital Advantage

- Key Insight
 - Is not just about convenience
 - Has built-in mechanisms that naturally lead to more accurate data
- Why?
 - **Expert Source:** Customers key in their own data
 - **Memory Aid:** Data is stored for later use
 - **Single Source of Truth:** Data is reused throughout the order fulfillment process
 - **Instant Proofreading:** Information is used as feedback to the customer

Customer-Driven Data: Accuracy Begins at the Source

- **Expert Source**

- **Core Idea:** Who knows your address and name better than you?
- **Reduce Transmission Errors**
 - **Traditional:** Ordering pizza over a phone line → “Was that ‘G-e-o-f-f’ or ‘J-e-f-f’?”
 - **E-commerce:** You type your own name and address, ensuring it is spelled correctly

- **Memory Aid**

- **Code Idea:** Good UX design makes returning easy and accurate
- **Autocomplete**
 - The browser/site suggests **matches as you type**
 - Saves keystrokes and prevents typos
- **Cookies**
 - Small, encrypted files stored on a **user's computer by a website**
 - Allows sites to remember your login, stored addresses, and payment info for next time
 - Only the company that **placed the cookie can access its information**

Reusing Data and Closing the Feedback Loop

- **Single Source of Truth**

- A single, accurate entry
- **Step 1:** Customer places an order by filling up all the information
- **Step 2:** That same data is used to:
 - Send an invoice
 - Tell the warehouse which product to pick
 - Print the shipping label
 - Send a tracking email
 - Update inventory levels

- **Instant Proofreading**

- Feedback turns the customer into a real-time data validator
- **Feedback Loop**
 - Customer accidentally orders 2 pairs of jeans instead of 1
 - Immediately receive an email confirmation: "Thank you for your order of 2 pairs of jeans"
 - The customer notices the mistake
 - They back to the site and **correct the order before it ships**